Certificate-Gated Neural Proposers for Verifier-Grounded Action Scoring

PLACEHOLDER: Authors and affiliations

Working draft — ideal-paper roadmap

Abstract. When an exact verifier is available, the right architecture for action scoring is not an end-to-end neural policy. We study a hybrid pipeline in which a deterministic certificate-checked gate prunes structurally impossible candidates before a small learned scorer ranks the residual. On a prime-gap benchmark with residue-sieve features removed from the scorer input, the raw scorer reaches 9.13% top-1; a fixed hand-wheel gate raises this to 45.12%, and a learned rule gate reaches 68.87% at 100% top-10 with an ≈6× reduction in model latency. The symbolic gate itself is produced by a neural proposer trained against a soundness-certificate verifier, which accepts 148 rules with zero false rejections out of 25,122 proposed candidates and recovers 69.40% of wheel-equivalent verifier work. Under sparse-exact-reward policy gradient on the same task, strict verifier RL inflates logit scale (ECE 0.070 → 0.127) without improving ranking; a KL-toward-prior + entropy regularizer keeps ECE at 0.060 at the same top-k. Prime gaps are used as a substrate, not the goal; we argue the protocol is transferable to any exact-verifier domain.

1. Introduction

Exact verifiers are common in domains where action correctness can be checked algorithmically even when discovery cannot: theorem proving, program synthesis, combinatorial search, formal shortest-path problems. The dominant style in modern learned policy work treats the verifier as a reward signal for an end-to-end neural policy. We argue this is the wrong default when the candidate space contains many structurally impossible actions that the verifier can cheaply eliminate.

Our thesis is the **certificate-gating loop**: a learned scorer should consume the **survivors** of a verifier-checked symbolic filter, and the filter itself can be produced by a neural proposer gated by a soundness-certificate verifier. The scorer ranks the residual; the gate eliminates the rest; nothing the gate emits can violate correctness because every rule it accepts carries a deterministic proof.

We make three claims, each tied to a concrete experiment in the released artifact:

(C1) **Certificate-gating dominates raw scoring.** On the canonical prime-gap benchmark, the minimal-feature raw scorer reaches 9.13% top-1. The same scorer reaches 45.12% under a fixed hand-wheel gate and **68.87% top-1 under a learned rule gate** at 100% top-10. The dominating effect is the gate, not the depth of the scorer.

(C2) **A certificate-gated proposer discovers sound symbolic gates.** A neural proposer over a small rule grammar, paired with a soundness-certificate verifier, produces 148 sound rules with zero false rejections, recovering 69.40% of wheel-equivalent verifier work. The soundness guarantee is a property of the certificate, not of training.

(C3) **KL-regularized verifier RL prevents calibration collapse.** Strict policy gradient on sparse exact rewards inflates logit scale rather than improving ranking; a standard KL-to-prior + entropy term restores calibration at unchanged top-k.

We use prime gaps as the experiment domain because they offer a fast deterministic verifier at all scales, a compact totally-ordered candidate space, an exhausted classical literature, and rich algebraic structure for the rule grammar. **We are not claiming improved prime generation.**

2. Related Work

PLACEHOLDER — Required citations and 1-paragraph positioning against each:

- Process supervision (Lightman et al., 2023): we use a similar verifier-as-step-checker pattern; their reward model is learned, ours is exact-by-construction.
- Verifier-guided decoding / best-of-N (Cobbe et al., 2021; Hosseini et al., 2024): we gate **before** scoring, not after.
- AlphaProof / AlphaGeometry symbolic-neural search (Trinh et al., 2024): our rule synthesis is structurally similar but verifier-grounded by certificate rather than by proof completion.
- RLHF KL-regularized policy gradient (Stiennon et al., 2020; Ouyang et al., 2022): our Finding 3 is this objective applied to exact-verifier rewards.
- Neural-symbolic program synthesis (NeSy literature).

3. Method

3.1. The Certificate-Gating Loop

The reusable architecture is:

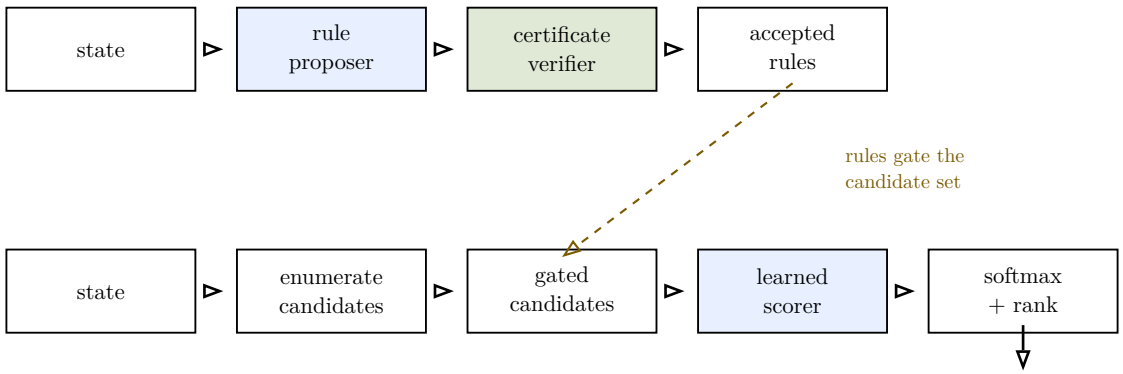


Figure 1: The certificate-gating loop. A learned rule proposer (top) produces candidate rules that a soundness-certificate verifier accepts or rejects. Accepted rules deterministically prune the candidate set; a shared neural scorer ranks the survivors; the exact verifier supplies the correctness signal.

The design invariant is **the model proposes, the verifier decides, the trace remembers.** A rule r is accepted only if a deterministic algebraic check certifies that r cannot reject the true action on any state in the support set, so soundness does not depend on the proposer being well-trained.

3.2. Trace and Checkpoint Contract

Every step writes a JSONL row with schema `crlp.trace.v1`, including a stable per-action identifier, the candidate scores and probabilities, the selected and target indices, the verifier result, and verifier cost. Replay resolves `target_index` → stable `action_id` → regenerated candidate row, so reordering action ids cannot silently change replay semantics. Checkpoints carry SHA-256 model identity, a `parent_model_hash`, and a `feature_encoder_version`; comparison reports (`crlp.compare.v1`) carry both hashes, the evaluated `feature_set`, and machine-readable reliability bins.

4. Experiment Domain: Prime Gaps

4.1. Setup

The state is a prime p; the candidate space is gaps $g \in \{2, 4, ..., 256\}$ plus an overflow action; the verifier returns the true next-prime gap. The default candidate encoding is the minimal feature set $[\log p, \text{norm_gap}, \text{overflow_flag}]$. A full-sieve feature set that adds precomputed divisibility flags for $p + g$ modulo small primes is retained only as a leakage ablation. The scorer is a shared MLP returning one scalar per candidate; softmax over candidates produces the proposal distribution.

The default split is $p < 800,000$ train, $800,000 \leq p < 1,000,000$ validation, $1,000,000 \leq p \leq 3,000,000$ test; all reported numbers use 30,000 held-out test states unless stated otherwise.

4.2. Why this domain

- A fast deterministic verifier at all scales (segmented sieves, wheel filters, block proofs).
- A compact totally-ordered candidate space.
- An exhausted classical literature — we cannot beat a sieve, which sharpens the question to "what does the learned component contribute under perfect feedback?"
- Rich algebraic structure for the rule grammar (residue classes, divisibility certificates).

5. Finding 1: Certificate-gating dominates raw scoring

The same trained scorer is evaluated under three gate conditions on the canonical 30,000-sample held-out benchmark. Top-1 results:

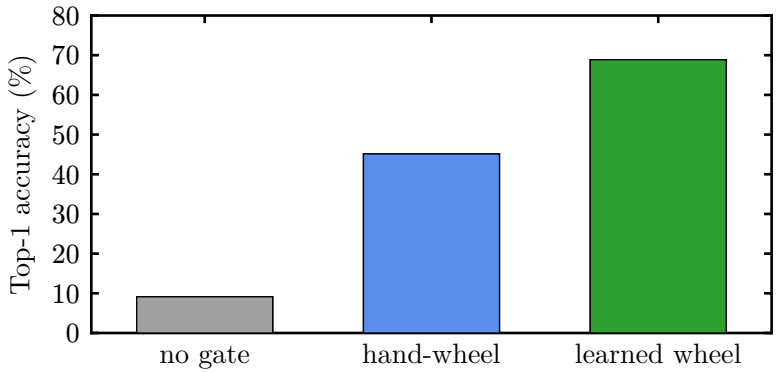


Figure 2: Top-1 accuracy by gate on the canonical 30,000-sample held-out benchmark. The clean raw-to-learned delta is **9.13 → 68.87**. The fixed-hand-wheel-to-learned delta is **45.12 → 68.87** (+23.7pp).

The former 45.12% raw-scorer result is now a full-sieve-feature leakage ablation, not the headline.

Full metrics:

Gate	Top-1	Top-10	NLL	ECE	ms/sample
no gate	9.13%	78.01%	2.9994	0.1076	0.0354
hand-wheel	45.12%	99.77%	1.5902	0.0834	0.0361
learned wheel	**68.87%**	**100.00%**	**1.0056**	0.1402	**0.0059**

Table 1: Gate ablation, minimal-feature scorer, identical weights across rows.

PLACEHOLDER — Required to elevate C1 from "load-bearing on prime-gap" to "general":

- **E1.1** — Random-rule gate at matched verifier-call budget: replace the learned rule set with k random sound rules and report the top-1 / top-10 / NLL curve as k varies. Resolves whether the **content** of the learned rules matters or only their count.
- **E1.2** — Replicate on a second domain. `grid_path` admits a natural certificate gate (legality + admissible-heuristic pruning). Same scorer + three gates + same table.
- **E1.3** — Investigate the learned-wheel calibration tradeoff. The learned wheel improves ranking sharply but has higher ECE than raw minimal and hand-wheel rows in the default run.
- **E1.4** — Confidence intervals across ≥ 5 seeds for the 45.12 → 68.87 delta.

6. Finding 2: A certificate-gated proposer discovers sound symbolic gates

The rule grammar is `if p % m == a then reject gap % m == b`; the certificate verifier accepts a rule iff $\gcd(a+b, m) > 1$ (so $p+g$ is necessarily composite) and the support count exceeds a threshold. Zero false rejections is a structural property of the certificate.

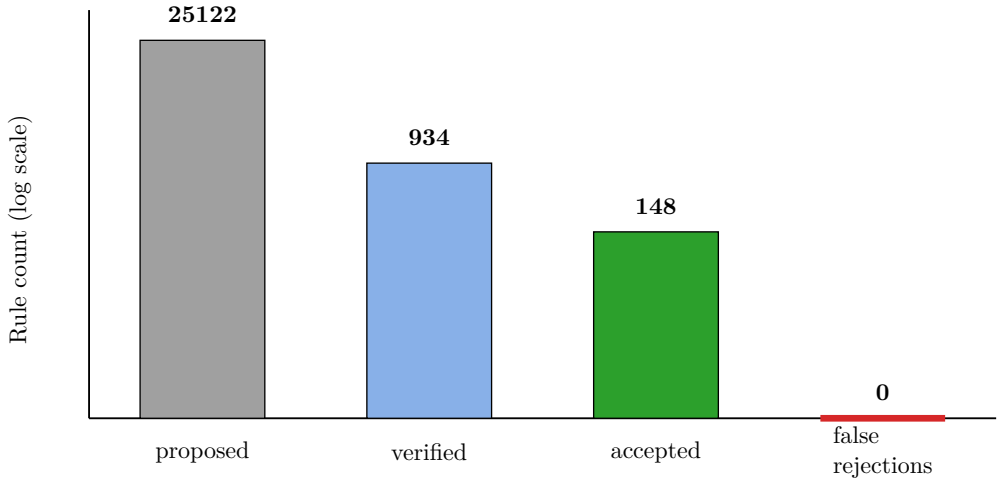


Figure 3: Rule-search acceptance funnel. Of 25,122 proposed rules, 934 pass the soundness certificate, 148 survive support and false-rejection screens, and **zero** falsely reject the true next-prime gap. The accepted set recovers 69.40% of wheel-equivalent verifier work.

What this evidence supports. "A certificate-gated proposer over this grammar produces a sound, useful rule set." The soundness comes from the certificate; the coverage comes from the combination of proposer ranking and enumeration.

What this evidence does not yet support. "The neural proposer, specifically, is essential." The numbers above are consistent with both "the proposer ranked rules well" and "a near-random proposer + the same verifier reaches the same rule set." Distinguishing the two requires E2.1 below.

PLACEHOLDER — Required to elevate C2 from "the verifier is doing useful work" to "the learned proposer is essential":

- **E2.1** — Random-rule-proposer baseline at fixed verifier-call budget. Sample rules uniformly from the grammar; run the same certificate verifier; report accepted rules and verifier work saved as a function of budget. The learned proposer's claim is then "matches the random baseline at $\approx N$x lower budget" for some defensible N.
- **E2.2** — Exhaustive enumerator at the same budget. For small enough m, exhaustive enumeration is feasible and lower-bounds the achievable coverage.

- **E2.3** — Generalize the grammar: extend to conjunctive rules (`if p % m == a and ... then reject ...`), report whether the proposer scales.
- **E2.4** — Show one **worked example** of an accepted rule the hand-wheel does not capture and explain why it generalizes.

7. Finding 3: KL-regularized verifier RL prevents calibration collapse

Starting from a single supervised checkpoint, we clone two RL branches: strict policy gradient on exact rewards, and the same gradient regularized with KL-to-supervised-prior + entropy.

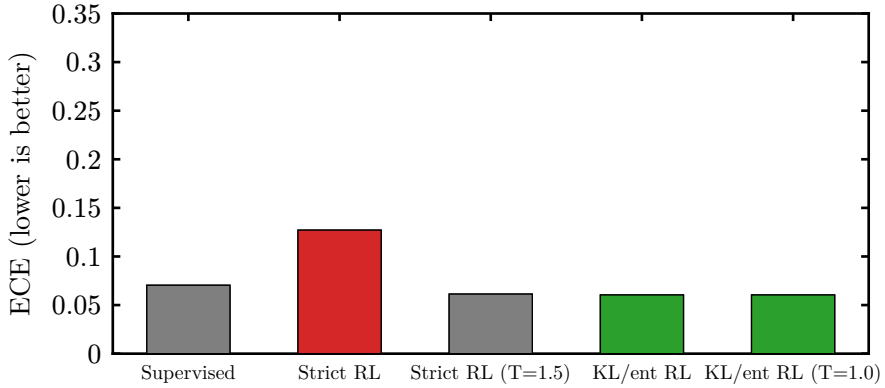


Figure 4: Expected calibration error by RL variant. Strict verifier RL degrades calibration (ECE 0.070 $\rightarrow$ 0.127) while leaving top-1 **unchanged at 9.13%** across all variants. Post-hoc temperature scaling needs $T = 1.5$ to recover; KL/entropy regularization fixes it at training time without that crutch.

Policy	T	Top-1	Top-10	NLL	ECE
Supervised	1.00	9.13%	78.01%	2.9170	0.0704
Strict verifier RL	1.00	9.13%	78.01%	3.0611	**0.1271**
Strict RL, temp-scaled	1.50	9.13%	78.01%	2.9063	0.0613
KL/entropy RL	1.00	9.13%	78.01%	2.9056	0.0604
KL/entropy RL, temp-scaled	1.00	9.13%	78.01%	2.9056	0.0604

Table 2: RL ablation, single cloned supervised checkpoint, default split.

The mechanism: strict policy gradient on a sparse exact reward signal pushes logit magnitudes without changing their relative ordering. The KL penalty bounds movement away from the reference policy; the entropy bonus discourages collapse to a single overconfident action. Together they reproduce the standard RLHF training stabilization, applied here to verifier rewards instead of preference rewards.

7.1. Reliability Diagram

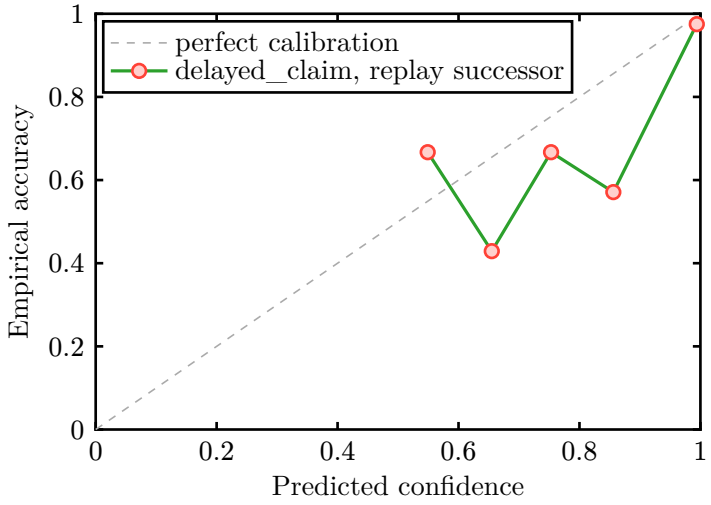


Figure 5: Reliability diagram for the trace-replay successor on `delayed_claim`. Bins from `results/latest/comparison.jsonl`. The high-confidence bin (894 of 1000 samples, $\hat{p} = 0.994$, $\hat{p}_{\text{emp}} = 0.975$) carries most of the mass; the low-population mid bins are noisier.

PLACEHOLDER: Replace with the prime-gap reliability diagram for supervised, strict-RL, and KL-RL once `make prime-gap` is run with `reliability_bins` capture enabled.

PLACEHOLDER — Required to elevate C3 from "true on prime-gap" to "a property of sparse-exact-reward RL":

- **E3.1** — Run strict-vs-regularized RL on `grid_path` and `delayed_claim` with the same protocol; report the analogous Table 2 for each.
- **E3.2** — Sweep the KL coefficient and entropy weight on a coarse grid for each task, plot the calibration-ranking Pareto frontier.
- **E3.3** — Per-bin reliability diagrams for **strict-RL vs KL-RL on prime-gap** (the current diagram uses delayed-claim as a stand-in because that's the only task whose `reliability_bins` are currently captured).
- **E3.4** — Confidence intervals across ≥ 5 seeds for the strict-vs-KL ECE delta.

8. Substrate

The hybrid architecture is the contribution; the substrate is the discipline that makes the claims auditable. Three pieces matter and would be a brief methods section in a longer paper:

- **Stable action IDs.** Trace rows carry `schema: "crlp.trace.v1"`, `trace_version: 1`, and `action_id_contract: "stable_unique_action_ids"`. Replay resolves `target_index` to the trace row's `action_id`, then maps that stable id back into the regenerated candidate row.
- **Checkpoint identity.** SHA-256 over the exact checkpoint bytes, plus `parent_model_hash` and `feature_encoder_version` in metadata. Successor models declare their parent; comparison reports carry both hashes and the evaluated `feature_set`.
- **Pure C, no third-party deps.** The neural library is dense layers, sigmoid/ReLU/tanh/linear, MSE and softmax cross-entropy, SGD, batched 4-by-4 tiled forward/backward, cloned networks, action scoring, pairwise ranking loss, and cached policy-gradient updates with optional KL/entropy regularization. ≈3k lines for the core; the prime-gap task is split into focused C modules for data, features, rule search, training, evaluation, and replay.

9. Limitations

L1. **Random-proposer baseline missing (Finding 2).** Until E2.1 lands, the load-bearing claim is "certificate-gated rule discovery is sound and recovers most of the wheel," not "the learned proposer is essential."

L2. **Cross-task verifier-RL comparison missing (Finding 3).** The pipeline supports it; the comparison is not in canonical results yet.

L3. **Single-domain headline (Finding 1).** Hybrid-wins is demonstrated on prime gaps only.

L4. **Statistical rigor.** Default `make prime-gap-experiments` runs 1 seed; the harness supports more. Deltas need confidence intervals across ≥ 5 seeds.

L5. **Learned-wheel calibration tradeoff** needs a reliability-diagram pass.

L6. **No literature comparison** on the calibration-collapse mechanism yet.

10. Conclusion

PLACEHOLDER — Write the conclusion last. Three sentences max: (a) certificate-gating is the architectural claim, (b) verifier-grounded rule synthesis is the constructive claim, (c) KL-regularized verifier RL is the training-stability claim. Avoid claiming the protocol scales; that is future work, not a result.

11. Roadmap to the ideal paper

This appendix tracks every experiment the current draft asks for but does not yet have. The goal is that every **PLACEHOLDER: ...** marker in the body resolves to one of these rows.

ID	Experiment	Effort	Defends
E1.1	Random-rule gate at matched verifier-call budget	2d	C1
E1.2	Replicate Finding 1 on `grid_path` with admissible-heuristic gate	3-4d	C1
E1.3	Investigate learned-wheel calibration tradeoff	1d	C1
E1.4	CIs across ≥ 5 seeds for 45.12 $\rightarrow$ 68.87 delta	0.5d	C1
E2.1	Random-rule-proposer baseline at fixed budget	2d	C2
E2.2	Exhaustive rule enumerator at same budget (where feasible)	1d	C2
E2.3	Conjunctive rule grammar extension	3-5d	C2
E2.4	Worked example of an accepted rule the hand-wheel misses	0.5d	C2
E3.1	Strict-vs-regularized RL on `grid_path` and `delayed_claim`	2-3d	C3
E3.2	KL/entropy Pareto sweep per task	2d	C3
E3.3	Reliability diagrams for strict-RL and KL-RL on prime-gap	0.5d	C3
E3.4	CIs across ≥ 5 seeds for strict-vs-KL ECE delta	0.5d	C3
W1	Related-work positioning paragraphs (see §2)	1d	all
W2	Conclusion + abstract revision after experiments	0.5d	all

Table 3: Roadmap. Total estimated effort: $\approx$3 weeks of focused work to take the current draft from "defensible as an arxiv preprint" to "submission-ready with all headline claims independently supported."

Minimum viable submission. E2.1, E1.3, E3.3, W1, W2. ≈5 days. Leaves Findings 2 and 3 with qualified scope (prime-gap only) but every claim in the body has a defensible experiment behind it.

Strong submission. Add E1.2, E2.4, E3.1, E1.4, E3.4. ≈2 weeks. All three findings have either a cross-task replication or a baseline strong enough to rule out the obvious alternative explanation.

Full ideal paper. Everything above. ≈3 weeks. The grammar extension (E2.3) is the optional ambition; if conjunctive rules generalize, the paper grows a fourth finding.